

Motherhood Penalties and Labor Informality in Latin America

Motherhood Wage Penalties in Latin America: The Significance of Labor InformalityAida Villanueva*, *University of Texas at Austin*Ken-Hou Lin, *University of Texas at Austin*

Previous research has established the presence of a motherhood wage penalty in many developed societies; however, whether mothers face similar disadvantages in developing countries remains underexplored. This article argues that different intervening factors emerge when considering mothers' labor compensation in developing contexts. Labor informality, a key characteristic of labor markets in developing countries, could play a significant role in shaping the wage consequence of motherhood. Using microdata from 43 national household surveys conducted between 2000 and 2017, we analyze five Latin American countries: Argentina, Brazil, Chile, Mexico, and Peru. After accounting for selection into employment and human capital, we find that mothers receive lower wages than childless women in all five countries. The penalties are similar to those found in some developed countries, ranging from 12 percent in Brazil to 21 percent in Chile. Mothers' higher likelihood to work in the informal sector accounts for part of the wage gap.

Keywords: motherhood penalty; labor informality; precarious work; Latin America.

Introduction

In recent decades, social science literature has demonstrated that many working mothers face a wage disadvantage compared to childless women (Budig and England 2001; Correll et al. 2007; Budig et al. 2012; England et al. 2016). Yet, motherhood penalties are neither universal nor similar across societies. For instance, Budig and colleagues' analysis of 22 countries finds penalties

Corresponding author aidavm@utexas.edu

This research was supported by grants, R24HD042849 and T32HD007081, awarded to the Population Research Center by the Eunice Kennedy Shriver National Institute of Child Health and Human Development, and by Brain Roberts, the C.B. Smith Sr. Chair in US-Mexico Relations at the University of Texas at Austin. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. We thank Lissette Aliaga, Jose Carlos Aguilar, Marta Ascherio, Celia Hubert, Paige Gabriel, Maria Laura Oliveri, David Pedulla, and Kelly Raley for their contributions to this paper.

ranging from 33 percent of mothers' annual earnings in West Germany to 0 percent in countries such as Australia, Belgium, and Finland (2012). While scholars have made significant progress toward understanding motherhood penalties in developed countries, we know far less about the association between motherhood and labor earnings in developing societies.

Moreover, the factors impacting mothers' labor compensation in developing countries may not be identical to those operating in developed societies. For instance, scholars investigating developed countries have underscored the importance of family-friendly policies in explaining women's labor outcomes and pay gaps affecting mothers (Misra and Strader 2013; Abendroth et al. 2014; Budig et al. 2016). Yet, developing countries often lack the capacity to enforce family-friendly regulations even when they are in place. Thus, these policies may have a weaker influence over women's pay outside of developed societies.

To further complicate the issue, informal work is a predominant feature of labor markets in developing economies and has important consequences for women's labor trajectories. Informal jobs are not properly regulated and commonly unregistered to governmental agencies. They often come without a signed contract, and, consequently, they lack protections against health hazards or unemployment shocks. While informal work is common among Latin American workers with low levels of education, a sizeable share of skilled workers in the region is informal as well (ILO 2014).

Women are overrepresented in the informal sector, and the factors that influence this pattern are gendered. While more stable and offering some protections, formal work in developing countries often requires inflexible schedules and working hours (Cerrutti 2000; Heymann 2006). Given women shoulder most of reproductive labor, they, and particularly mothers, are likely pushed out of the formal sector due to their need to combine paid work with family responsibilities. Importantly, the prevalence of informal work among Latin American women means that several nominal policies targeted to working mothers are not applicable to substantial segments of them.

In sum, despite the large body of literature on the motherhood wage penalty accumulated in the past decades, we know rather little about wage gaps affecting mothers in developing countries, and the factors that may explain these mothers' disadvantages are likely to be distinct from those identified for developed countries. We investigate motherhood pay gaps in five Latin American countries: Argentina, Brazil, Chile, Mexico, and Peru. Moreover, we explore the interplay between motherhood wage gaps and informal work, the defining characteristic of labor markets in developing countries. We hypothesize that sorting into precarious work could be a critical mechanism through which mothers are disadvantaged.

This study takes advantage of the richest and most-recent data available from this region: the primary population-representative household survey for each country. We harmonize 43 waves of data collected between 2000 and 2017. Taken together, our sample represents approximately 67 percent of the Latin American population. The countries we selected offer substantial variation in

key parameters such as level of labor informality and female labor force participation, which allows us to detect whether the motherhood pay gap is sensitive to national contexts. Overall, the results point to a substantial wage difference affecting mothers vis-à-vis childless women in Latin American countries. We also find that mothers' greater likelihood of working in the informal sector explains part of the disadvantage.

Although we focus on Latin American countries, the concept of labor informality resonates with that of precarious work in developed societies. Where informal jobs remain rare in developed countries, an increasing number of formal sector jobs are becoming precarious, characterized by lower wages and higher job insecurity (Kalleberg 2011). In the United States, precarious jobs are overwhelmingly occupied by women, and especially mothers. Our analysis of how precarious jobs affect working mothers in developing countries, therefore, has broader implications for ongoing discussions about gender wage gaps in developed societies.

The Motherhood Wage Penalty

Main explanations for the wage gap attributable to motherhood fall into two broad perspectives, concerning either labor market demand or supply. Demand-side explanations highlight the role of discrimination against mothers. Given motherhood is a devalued status in the workplace, employers tend to perceive mothers as less capable or less committed, regardless of their actual productivity (Correll et al. 2007; Benard and Correll 2010).

Supply-side theories focus instead on individual and household characteristics, noting differences between mothers and nonmothers in human capital investments, skill, and work-related preferences. Since mothers commonly shoulder the bulk of unpaid family labor (Killewald and Garcia-Mangano 2016), they may have less energy to invest in their employment (Kuhhirt and Ludwig 2012). Thus, mothers might be less productive than childless women, which could explain the disparities in pay.

Changes in work trajectories associated with motherhood, such as reducing hours or self-employment, are often placed under the umbrella of supply-side mechanisms. However, scholars have questioned whether such transits should be considered voluntary. Overall notions of standard employment are based on a gendered concept of an ideal worker, who has no household obligations (Acker 1990). While mothers in developed societies often describe their transitions to precarious forms of work using the language of "choice," a closer look unveils instances of structural constraints and discrimination (Stone 2007; Webber and Williams 2008).

In this spirit, our study aims to estimate a wage impact of motherhood on women, which includes the effect of mediators such as reduced work experience and occupational sorting, since these factors are understood as part of the causal pathway explaining mothers' labor compensation. In other words, we view interruptions in work trajectories as a common consequence of parenting for

women, to the detriment of mothers' wages (England et al. 2016). Then, an overall or total wage gap associated to motherhood is obtained after controlling for factors preceding or plausibly exogenous to parenting (e.g., education or race). Our study provides this type of estimate for a set of Latin American countries. In contrast, a net effect of motherhood would be obtained after adjusting for factors such as work experience, part-time employment, and tenure. Estimates on a net penalty could inform whether motherhood affects wages even among those with the same work experience, perhaps due to discrimination or lower productivity among mothers (England et al. 2016). In ancillary analysis, we present estimates after accounting for some endogenous factors.¹ While not the focus of this study, this set of ancillary analysis accounts for occupation, part-time employment and tenure; the latter as a proxy for work experience.

Mothers' Labor Compensation in Latin America

While empirical analyses on mothers' labor compensation in Latin America are scarce, various accounts point to mothers being disadvantaged from both a supply and demand perspectives. Discriminatory practices against working mothers are common, even during pregnancy and despite labor regulations that prohibit them (Cerrutti 2000; Heymann 2006; Frías 2011). For instance, Ansoleaga et al. (2011) find that pregnant women in Chile are frequently harassed by employers, in the worst-case scenarios, with the intention to pressure pregnant women into leaving their jobs. Supply-side factors also pose obstacles to mothers' paid work. Studies on time use report that responsibilities for household labor remain strongly gendered across the region. Thus, for each additional child, Brazilian and Peruvian mothers' time spent in family work increases, while fathers' stays constant (Freyre and Lopez 2011; Pinheiro et al. 2016).

Furthermore, work-family policies in Latin America, although limitedly enforced, are aligned to traditional gender roles. Regulations on parental leaves are overwhelmingly targeted at mothers², while access to subsidized childcare remains low. Day-school hours are also low in most of the region (Holland et al. 2015), which means less time available for mothers to work for pay.

Taken as a whole, one would expect a sizeable wage gap between mothers and childless women in Latin America, plausibly larger than those in developed countries. However, existing studies did not find consistent evidence of pay gaps affecting mothers. Piras and Ripani (2005) conducted a pioneer study with data collected around 1999 from Bolivia, Brazil, Ecuador, and Peru. They found a penalty affecting mothers in Peru; no penalty for Ecuador and a premium for mothers in Bolivia and Brazil. Gamboa and Zuluaga (2013) examined the motherhood penalty for Colombia, finding that the wage gap affecting mothers could be entirely explained by other observed characteristics. These two studies included very-young women in their analysis, which could have resulted on a lower penalty identified, since young women are likely to be both childless and

receive low wages. Furthermore, using data from Argentina, [Casal and Barham \(2013\)](#) provided a careful account of the motherhood penalty and considered the role of the informal sector in explaining mothers' disadvantages. They find a penalty concentrated among mothers in the informal sector. Yet, the authors used a particularly restrictive definition of formal work, pertinent for the case of Argentina, but that could have resulted in formal workers being a highly selected group.

Our analysis builds upon this body of research and aims to provide an assessment of the presence and magnitude of motherhood wage gaps in Latin America while taking the role of informal labor into account. To ensure that we compare mothers and nonmothers who are otherwise similar, we limit the analyses to women of prime childbearing age and consider differential selection into employment. To discern the broader patterns in Latin America, we analyze data from five large countries in the region: Argentina, Brazil, Chile, Mexico, and Peru. Considering the mounting evidence of gender and parenthood-based workplace discrimination in Latin America, we begin our analysis with the following expectation:

Hypothesis 1: Otherwise equal, mothers receive lower wages than childless women in Latin America.

Linking the Motherhood Penalty and Labor Informality

A large informal sector is a defining feature of labor markets in developing countries, with important implications for the structure and characteristics of employment and earnings ([Perry, 2007](#)). With about a half of the labor force working in the informal sector, unregulated work remains a policy priority across Latin America. Commonly, informal workers engage in employment relationships without legally recognized contracts and, therefore, they cannot be traced through administrative records. Their jobs do not offer protections against health or unemployment shocks, nor savings toward public pension. Common characteristics of informal work include lower wages, greater instability, easy entry, and lack of benefits ([Cerrutti 2000](#); [Villarreal and Blanchard 2013](#)). Typical jobs in the informal sector include work in construction, domestic work, and work conducted in small firms. About 50 percent of informal jobs in Latin America are located in the sectors of construction, retails, restaurants, and hotels ([ILO 2014](#)).

At the same time, informal work exists across a wide range of industries. Unskilled labor is overwhelmingly informal; therefore, informality is correlated with characteristics that are negatively associated with wages. Yet, a significant proportion of skilled work is informal as well. In our analytical samples, the share of informal workers in professional and managerial occupations ranges from 5 percent in Chile to 31 percent in Mexico.

While labor informality is a heavily researched topic in Latin America, empirical contributions tend to focus on men. Women, however, are overrepresented among informal workers, and taking an informal job is more consequential to

women's career trajectories (UN Women, 2015; ILO 2014). Once working in the unregulated sector, women are less likely to move to formal work (Gong and van Soest 2002), while the wage gap associated with informality tends to be larger for women than men (Tornarolli et al. 2014). Moreover, prolonged years of informal work result in lower pensions at old age, since pension systems are tied to a formal job history (CEPAL 2018). Therefore, informality contributes to women's risk to fall into poverty at old age.

The processes behind women's overrepresentation in the informal sector are gendered. Informal work in developing countries often provides more flexibility in terms of schedules, working hours, and location of work than employment in the formal sector. Given women remain responsible for the bulk of family work, they often look for options that allow them to combine unpaid household labor with employment, and such options are more common in the informal sector (Cerrutti 2000; Heymann 2006; Cassirer and Addati 2007; Alfes 2016). Therefore, the benefits of informal work for mothers are tied to the gendered distribution of reproductive labor. In this vein, the role of informal labor at explaining mothers' wages is better understood as part of a broader structure of disadvantage affecting women in general and mothers in particular. In this study, we see informal work as a critical point in the path connecting motherhood to lower wages in Latin America.

While few empirical studies have focused on women and informal labor, researchers on labor markets in the region usually acknowledge that women's need for flexibility plays a role in their overrepresentation among informal workers (Piras 2004; Pages and Piras 2010; UN Women 2015). Moreover, a small body of literature considers women's transits to informal work in developing countries from a qualitative perspective. Heymann and colleagues document the experience of mothers who were unable to either get or retain a formal job due to their caregiving responsibilities (Heymann 2006). Women reported that formal jobs would not allow any sort of leave or change in schedule for taking care of a sick child, for instance. Under such circumstances, mothers often chose an informal, lower quality job that allowed them to take care of their children, either by working from home (as home-based garment workers, for example) or by keeping flexible schedules (e.g., domestic workers).

Since the informal sector provides more flexibility than the formal one, we believe mothers make first the decision to work in the informal sector and then look for specific jobs. That is, predominantly, mothers' decision to join the informal sector precedes the one on the specific job to take. Therefore, and given informal jobs are largely precarious, transitions to the informal sector are more significant to women's work trajectories than specific occupations and other work-related characteristics. Qualitative evidence supports this notion. In her description of work trajectories of women in Mexico City and Buenos Aires, Cerrutti explains that most of low-socioeconomic-status women interviewed engaged in "informal activities," such as domestic workers or manual workers in workshops. The provision of flexibility was key to these decisions (Cerrutti 2000).

We argue that accounting for informal work is a key to understanding women's participation in the workforce in Latin America, and therefore an important component for studying motherhood penalties. In the light of the above mentioned, we expect that mothers will be more likely to engage in informal work, all else equal:

Hypothesis 2: Mothers in Latin America are more likely to work in the informal sector than childless women.

Moreover, given informal work tends to pay lower wages, we anticipate that mothers' greater likelihood to have informal jobs will contribute to overall motherhood pay gaps.

Hypothesis 3: The motherhood wage penalty in Latin America is explained, at least in part, by labor informality.

Figure 1 shows proportions of informality among salaried workers in 16 Latin American countries. This study includes some of the countries with the higher rates of informal labor in the region, such as Mexico and Peru (66 and 55 percent in 2012), as well as Chile, which has the lowest share of informal workers (18 percent). Rates for Brazil and Argentina lie somewhat in between.

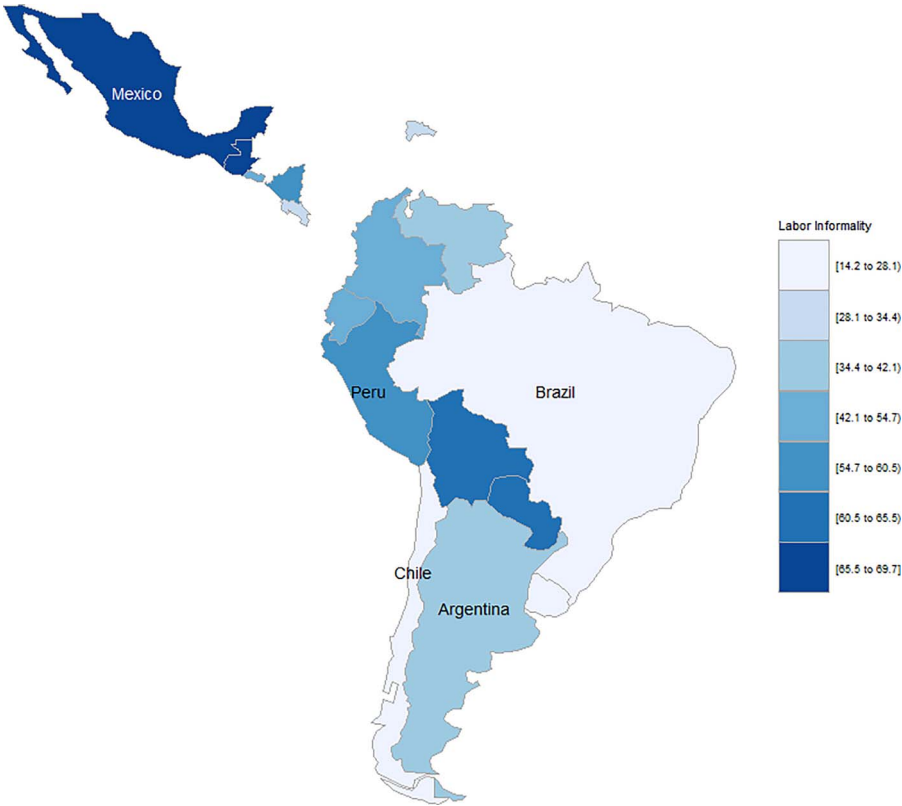
Research Design

Data and Sample

We draw on the latest data from the primary population-representative household survey for each country. All surveys were conducted by national census bureaus and are fully probabilistic, multistage, and stratified. We work with the following country years: Argentina (2004–2014), Brazil (2005–2015), Chile (2000–2015), Mexico (2008–2016), and Peru (2008–2017). See [Supplement 2](#) for additional information.

Our sample is limited to women aged 25–40 living in urban areas, who are either household heads or partners.³ The lower bound age restriction intends to exclude women who have not completed their formal education, and the upper-bound limit aims to minimize the number of mothers whose children no longer live with them (and would appear as childless in our data).⁴ We exclude zero-income family workers from our sample⁵, women who are self-employed and those living in rural areas. While self-employed mothers may also be disadvantaged, the mechanisms that generate the penalty are likely to differ from those in place for employed mothers. Lastly, in developing countries, women in rural settings tend to keep multiple production activities centered around the household, which makes the count of working hours a less meaningful measure (FAO 2010). Unfortunately, the exclusion of self-employed and rural workers results in a nontrivial reduction in our analytical samples, particularly for Peru.⁶ While we see the restrictions applied as pertinent, we conducted

Figure 1 Proportion of Workers in the Informal Sector by Latin American country, 2010s.



Source: Socioeconomic Database for Latin America and the Caribbean (Universidad de la Plata and World Bank). The Indicator of Labor Informality Shown is the Same Used in this Study.

supplemental analyses including rural workers and self-employed women (Supplement 14). Results do not depart drastically from estimates shown in the main text.

Descriptive Statistics

Table 1 presents key socioeconomic indicators for the countries included in our study, at the country level. The countries selected not only share important similarities but also present substantial variation on important characteristics. First, their employment structure points to similar levels of industrialization. Across countries, we see a predominance of the services sector—from 62 percent of the workforce in Mexico to 77 percent in Brazil. Second, after decades of internal migration, these countries are largely urban. Furthermore, the estimated number of children per woman converges around replacement rate. It is worth noting that, since Table 1 presents information at the country level, it shows a

picture that is more disadvantaged than that present in our analytical samples, given the exclusion of rural areas from our analyses.

Table 1 also shows important differences across the countries examined. Argentina and Chile have a lower proportion of population living at \$3.10 or less per day, while rural populations remain above 20 percent only in Peru and Mexico. Furthermore, female labor force participation ranges from 43 percent in Mexico to 72 percent in Peru. Traditionally, Chile and Mexico present some of the lower rates of female participation in the workforce in the Latin American region.

Table 2 shows additional indicators, drawn from the data sources we use in this study. We present means and proportions for working women who are either household heads or partners, mothers and childless, 25–40 years old, in urban areas. In all countries, mothers earn less than nonmothers. Compared to mothers, childless women are more likely to have some level of tertiary education and to work in professional or managerial occupations. Interestingly, the proportion of women working 55 and more hours per week does not differ importantly between mothers and nonmothers. Across all countries, mothers are overrepresented in the informal sector.

Measures

Dependent variable

The outcome of interest is the logged hourly earnings from the respondents' main job⁷ (Budig and England 2001; Gangl and Ziefle 2009; Killewald 2013), where earnings are expressed in US dollars by transforming them from local currency.⁸ In Latin American surveys, questions on labor earnings are often asked in reference to a month period. We calculate hourly wages as earnings corresponding to the period of reference, divided by reported hours worked per week, and including all sources of compensation additional to the base wage or salary.⁹ We top-code weekly hours at 105 or more and exclude workers who did not report hours of work.

Key independent variables

Motherhood is measured with a dichotomous variable indicating the presence of any children (ages 0–18) in her family. We use alternative measures such as the number of children and age of children in supplementary analyses (Supplements 15 and 16). Informality is operationalized by identifying whether the worker is affiliated to a contributory pension system, which gives her the right to receive a pension when retired (Villarreal and Blanchard 2013; Tornarolli et al. 2014). In Latin America, this contribution is part of a “basic package” of social security benefits that is commonly tied to formal employment. Although enrollment in this package does not assure access to every benefit mandated by law (including those associated with motherhood), this measure is the best proxy.

Table 1. Key Socioeconomic Indicators by Country, 2010s

	Argentina	Brazil	Chile	Mexico	Peru	Source
GDP per capita (current US\$)	13,432 (2015)	8,539 (2015)	13,416 (2015)	9,005 (2015)	6,027 (2015)	WDI, World Bank
Poverty headcount ratio at \$3.10 per day (2011 PPP)	4.28 (2014)	7.56 (2014)	2.05 (2013)	10.95 (2014)	9.01 (2014)	WDI, World Bank
Rural population	8.25% (2015)	14.31% (2015)	10.47% (2015)	20.75% (2015)	21.39% (2015)	WDI, World Bank
Female labor force participation (ILO estimate)	48%	54%	47%	43%	72%	WDI, World Bank; around 2010
<i>Educational outcomes</i>						
Mean years of schooling (females, ages 25 and older)	10	7.3	9.6	8.2	8	International Human Development Indicators (UN); data for 2010
Compulsory education, duration in years	13	9	12	11	12	WDI, World Bank; around 2010
Female population with at least some secondary education (% ages 25 and older)	56.3	51.9	73.3	51	53.2	International Human Development Indicators (UN); data for 2010
<i>Economic structure</i>						
Employment in agriculture	0.50% (2014)	14.5% (2013)	9.20% (2013)	13.40% (2013)	24.7% (2007)	WDI, World Bank
Employment in industry	24% (2014)	23% (2014)	24% (2013)	24% (2013)	23% (2014)	WDI, World Bank
Employment in services	75% (2014)	77% (2014)	67% (2013)	62% (2013)	76% (2014)	WDI, World Bank
Informal labor	34.2% (2013)	22.9% (2013)	15.5% (2013)	65.5% (2012)	53.9% (2013)	Socioeconomic database for LAC, Universidad de La Plata, and World Bank
Access to electricity	99.8% (2012)	99.5% (2012)	99.6% (2012)	99.1% (2012)	91.2% (2012)	WDI, World Bank
<i>Fertility trends</i>						
Fertility rates (births per woman)	2.3 (2015)	1.8 (2015)	1.8 (2015)	2.3 (2015)	2.2 (2015)	U.S. Census Bureau
Age at first birth	24.4 (2014)	21 (2006)	23 (2007)	21.1 (2014)	22.2 (2013)	Official statistics per country
Proportion of childless women, 41 and older ¹	11.02	12.25	8.91	7.41	5.13	Latest census available

Notes: Own calculations using census data provided through IPUMS.

Table 2. Means and Proportions. Working Women, Only Salaried. Mothers and Childless, 25–40 Years Old, Household Heads and Partners, in Urban Areas

	Argentina, 2000–2014		Brazil, 2005–2015		Chile, 2000–2015		Mexico, 2008–2016		Peru, 2008–2017	
	Mothers	Childless	Mothers	Childless	Mothers	Childless	Mothers	Childless	Mothers	Childless
Childless		0.29		0.24		0.20		0.16		0.15
Informal worker	0.44	0.26	0.27	0.17	0.19	0.13	0.59	0.46	0.55	0.41
Age	33.98 (4.15)	31.33 (4.32)	33.40 (4.30)	31.74 (4.67)	34.01 (4.31)	31.57 (4.66)	33.96 (4.33)	32.30 (4.87)	34.11 (4.31)	32.27 (4.71)
Has a partner	0.80	0.59	0.79	0.70	0.75	0.65	0.79	0.61	0.78	0.47
Minority	0.05	0.04	0.50	0.43	0.07	0.05	0.03	0.03	0.13	0.10
<i>Ed. attainment.</i>										
Elementary & less	0.20	0.06	0.26	0.12	0.12	0.05	0.24	0.12	0.16	0.07
H.S. incomplete	0.16	0.06	0.10	0.06	0.11	0.05	0.34	0.19	0.14	0.07
H.S. complete	0.23	0.16	0.41	0.40	0.39	0.22	0.20	0.16	0.28	0.17
Some tertiary & more	0.41	0.72	0.22	0.42	0.38	0.69	0.22	0.53	0.42	0.69
Hourly wage US \$	3.48 (3.19)	4.13 (3.19)	2.96 (3.84)	4.21 (5.13)	3.60 (4.10)	5.55 (5.31)	2.39 (2.64)	3.33 (3.06)	2.18 (2.57)	3.19 (3.73)

(Continued)

Table 2. Continued

Argentina, 2000–2014		Brazil, 2005–2015		Chile, 2000–2015		Mexico, 2008–2016		Peru, 2008–2017	
Mothers	Childless	Mothers	Childless	Mothers	Childless	Mothers	Childless	Mothers	Childless
<i>Occupation</i>									
Elementary workers	0.32	0.14	0.29	0.15	0.22	0.29	0.15	0.38	0.20
Plant/machine operators/craft/trade workers	0.06	0.04	0.08	0.06	0.04	0.10	0.06	0.07	0.05
Services and sales	0.17	0.14	0.25	0.23	0.27	0.25	0.20	0.19	0.16
Clerical	0.17	0.25	0.13	0.19	0.15	0.10	0.12	0.09	0.16
Technicians and associate prof	0.21	0.26	0.10	0.11	0.14	0.08	0.11	0.09	0.16
Professional & managerial	0.07	0.18	0.15	0.26	0.18	0.18	0.36	0.17	0.27
<i>Work intensity</i>									
Part time	0.61	0.38	0.26	0.19	0.20	0.34	0.16	0.37	0.23
55 & more hours	0.05	0.07	0.04	0.05	0.07	0.13	0.19	0.17	0.18
N	68,793	22,665	117,589	36,975	28,621	12,111	2,215	11,395	1,778

Notes: See Supplement 2 for complete list of sources.

Control variables

Measures for family characteristics include a dummy variable indicating marital status (married, cohabiting = 1). We also include a measure for ethnic, racial, or national minority. This captures indigenous ethnicity in Chile, Mexico and Peru, racial minority in Brazil, and labor immigrants in Argentina.¹⁰ Educational attainment is coded with a four-category measure, pertinent to each country and ranging up to “some tertiary education and more.” We further control for age and age squared. All models control for within-country region and survey year. [Supplement 5](#) shows additional information on the harmonization conducted.

We conducted ancillary analyses accounting for some of the factors located in the causal pathway explaining mothers’ disadvantage: occupation, part-time employment, and tenure in the current job. Where rather imperfect, tenure is included as a proxy for work experience¹¹. Thus, in this complementary set of analyses, we adjust out for part of the motherhood pay gap due to mediating factors such as occupational sorting and shorter tenures.

Analytical Strategy

Our first goal is to establish the presence and size of the motherhood pay gap across all five countries. To test Hypothesis 1, we estimate the wage penalty for motherhood with a two-stage Heckman equation:

$$\text{Ln}(W_{i,j}) = \beta_{0,j} + \beta_{1,j}M_{i,j} + \sum_k^K \beta_{k,j}X_{k,i,j} + \beta_{2,j}R_{i,j} + u_{i,j} \quad (1)$$

where i indexes individual in country j ; $M_{i,j}$ denotes her parental status; $X_{k,i,j}$, other characteristics: educational attainment, whether the respondent belongs to an ethnic, national, or racial minority, whether the respondent has a partner or not, age and age squared. All models control for region within country and survey year. The coefficient of interest is $\beta_{1,j}$, the conditional wage difference between mothers and childless women; based on Hypothesis 1, we expect it to be negative. $R_{i,j}$ denotes the inverse Mills ratio of being employed, which addresses the differential selection into employment. Following empirical literature on motherhood penalties, we include the following parameters in our selection model¹²: marital status, age, education, whether there are children 6 or younger in the household, and logged household income after subtracting women’s labor earnings ([Harkness and Waldfogel 2003](#); [Mandel and Semyonov 2005](#); [Budig et al. 2012](#)). Additionally, we included in the selection model a dichotomous variable capturing the presence of at least one female adult kin member, ages 19–64, who is not a household head or partner and may potentially share home responsibilities. The presence of a female relative in the household has been found positively associated to continued employment in Buenos Aires (Argentina) and Mexico city ([Cerrutti 2000](#)). With the above-noted covariates, we aim to identify an overall wage gap between mothers and childless women, as noted before, meaning that it *includes* factors plausibly in the causal pathway

explaining mothers' disadvantage in the labor markets, such as total work experience and occupational sorting (England et al. 2016).

Our second goal is to establish whether mothers are more likely to work in the informal sector (Hypothesis 2), which would suggest that mothers are pushed towards precarious jobs. To that end, we implement a logistic model predicting work in the informal sector (0/1) based on supply-side characteristics. We specify the model as:

$$\text{logit}(I_{i,j}) = \beta_{0,j} + \beta_{1,j}M_{i,j} + \sum_k^K X_{k,i,j} + \beta_{2,j}R_{i,j} + u_{i,j} \quad (2)$$

where $M_{i,j}$ indicates parental status, while $R_{i,j}$ the inverse Mills ratio of being employed, and $X_{k,i,j}$ represents relevant characteristics: age and age squared, race/ethnicity, educational attainment, region within country, and year.

Third, to test whether working in the informal sector contributes to the motherhood penalty (Hypothesis 3), we add to our first set of models an indicator of informality:

$$\text{Ln}(W_{i,j}) = \beta_{0,j} + \beta_{1,j}M_{i,j} + \beta_{2,j}I_{i,j} + \sum_k^K \beta_{k,j}X_{k,i,j} + \beta_{3,j}R_{i,j} + u_{i,j} \quad (3)$$

where $I_{i,j}$ captures if a woman is working in the informal sector and $\beta_{2,j}$ is expected to be negative. We then compare $\beta_{1,j}$ between Models (1) and (3) with a Wald Chi-Squared test to see if the difference in our coefficient for "mother" is statistically significant after accounting for informal labor.¹³ That is, we tested the null hypothesis that our coefficient for "mother" was the same across the two specifications: a first one without including informality and a second one controlling for this measure.

Ancillary Analyses

Accounting for occupation, part-time employment, and tenure

In complementary analysis, we re-estimated equation 3 by adding measures on occupation, reduced working hours, and work experience. Though our proxy for work experience (tenure) is rather imperfect, the latter set of estimates is intended to be closer to a "net effect" of motherhood, that is, accounting for factors that are endogenous to motherhood and earnings.

Matching mothers and childless women

Finally, we replicated equation 3 using matched samples of working women with different parental status but similar in other characteristics, which addresses the potential sample imbalance between mothers and nonmothers (Iacus et al. 2012; Pager and Pedulla 2015). Specifically, we exact-match mothers with nonmothers using a coarsened exact matching procedure (Iacus et al. 2012). It should be

noted that the matching does not eliminate omitted variable bias (see [Supplement 11](#) for additional information).

Results

Effects of Motherhood on Wages

[Table 3](#) presents the results from our first model, corrected for selection into employment with a Heckman procedure. Results using an Ordinary Least Squares (OLS) estimation are shown in [Supplement 9](#). Across all countries, results indicate a negative association between motherhood and wages. The wage gap ranges from 12 percent in Brazil to 21 percent in Chile. Furthermore, our estimates suggest that, on average, mothers earn 15 percent less than childless women in Argentina, 14 percent less in Mexico, and 20 percent less in Peru. While this range offers some variation, differences are not as substantial as those found for developed regions. Using a similar approach, Budig and colleagues find penalties in annual earnings ranging from 0 to 33 percent in a group of developed nations ([Budig et al. 2012](#)).¹⁴

Overall, these findings support Hypothesis 1 and suggest that a significant motherhood wage gap exists among Latin American countries. Other results in [Table 3](#) are aligned with previous studies. For all countries, we observe a penalty for ethnic, racial, or national minorities. We also find a strong educational premium, which is a traditional feature of Latin American labor markets and plays an important role in the high levels of income inequality in the region ([Torche 2014](#)).

Informal Work and Mothers' Labor Compensation

[Table 4](#) presents results for logistic models predicting work in the informal sector, with motherhood status as main independent variable. We see that, for all countries, our motherhood coefficient is positive, although only marginally so for the case of Chile (which has the smallest informal sector). That is, for all countries but Chile, results indicate that mothers are more likely to work in the informal sector compared to childless women, net of other relevant factors associated with labor supply, and with that association being both statistically significant and important in magnitude.

Other coefficients are aligned with the body of literature on informal labor in the region. With the exception of Mexico, the coefficient for year is negative. This is consistent with a well-documented reduction in labor informality in the region during the past decade, with Mexico being a notable exception ([Tornarolli et al. 2014](#)). Schooling is negatively associated with informal work for all countries analyzed, while minorities are more likely to hold informal jobs (although the coefficient fails to statistical significance for Chile). These results support our second hypothesis, suggesting that working mothers in Latin America are pushed into precarious forms of employment. Given the formal sector is characterized by rigid, "all-or-nothing" arrangements, mothers are likely to look for jobs in the informal sector. While largely precarious, informal jobs often allow women to

Table 3. Effect of Motherhood on Women’s Hourly Wages (ln), Models Controlling for Selection into the Labor Force

	Argentina, 2004–2014	Brazil, 2005–2015	Chile, 2000–2015	Mexico, 2008–2016	Peru, 2008–2017
	Heckman	Heckman	Heckman	Heckman	Heckman
Mother	–0.147*** (0.0151)	–0.118*** (0.00398)	–0.210*** (0.00922)	–0.143*** (0.0202)	–0.203*** (0.0176)
Age	0.130*** (0.0211)	0.108*** (0.00551)	0.140*** (0.0118)	0.118*** (0.0247)	0.0795*** (0.0208)
Age squared	–0.00141*** (0.000319)	–0.00136*** (0.0000836)	–0.00193*** (0.000178)	–0.00129*** (0.000372)	–0.000820** (0.000312)
Household head	–0.0483* (0.0213)	0.0796*** (0.00533)	0.304*** (0.0158)	0.355*** (0.0318)	0.211*** (0.0229)
Minority	–0.109*** (0.0278)	–0.107*** (0.00329)	–0.106*** (0.0137)	–0.229*** (0.0376)	–0.0840*** (0.0187)
Educ. Ref.: elementary & less ¹					
H.S. incomplete	0.0455* (0.0219)	0.176*** (0.00608)	0.150*** (0.0147)	0.316*** (0.0194)	0.119*** (0.0220)
H.S. complete	0.514*** (0.0214)	0.440*** (0.00501)	0.467*** (0.0134)	0.746*** (0.0227)	0.260*** (0.0192)
Some tertiary and more	1.014*** (0.0266)	1.338*** (0.00802)	1.480*** (0.0206)	1.626*** (0.0282)	1.013*** (0.0220)
Inverse Mills	0.216*** (0.0431)	0.352*** (0.0131)	0.693*** (0.0281)	0.569*** (0.0468)	0.530*** (0.0382)
Constant	–3.412*** (0.356)	–2.527*** (0.0924)	–3.188*** (0.198)	–2.803*** (0.414)	–2.397*** (0.346)
N	91,458	154,564	34,562	14,326	13,173

Notes: See [Supplement 2](#) for Complete List of Sources. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

combine paid work with family labor ([Cerrutti 2000](#); [Heymann 2006](#); [Cassirer and Addati 2007](#); [Alfers 2016](#)).

Table 5 illustrates results for Hypothesis 3. We compare our first specification (Model 1) with a second one that controls for informal work (Model 2), showing only estimates after adjusting for selection into the workforce (Heckman models). Thus, our coefficients for motherhood in Model 1 are the same shown in Table 3. As expected, labor informality is associated with lower wages in all five countries, net of other key characteristics explaining labor compensation.

We provide information on the percent change for the decrease between Model 1 and Model 2, as well as results for a Wald Chi-Squared test comparing our coefficient for “mother” across the two models. The wage gap due to

Table 4. Logistic Models Predicting Women's Work in the Informal Sector. Heckman Selection Models; Results in Odds Ratios

	Argentina, 2004–2014	Brazil, 2005–2015	Chile, 2000–2015	Mexico, 2008–2016	Peru, 2008–2017
	Heckman	Heckman	Heckman	Heckman	Heckman
Mother	1.560*** (0.0745)	1.200*** (0.0239)	1.168 ⁺ (0.104)	1.450*** (0.134)	1.445*** (0.123)
Age	0.724*** (0.0440)	0.942* (0.0246)	0.815* (0.0789)	0.654*** (0.0710)	0.741** (0.0729)
Age squared	1.004*** (0.000922)	1.001 (0.000396)	1.003 ⁺ (0.00146)	1.006*** (0.00165)	1.003* (0.00149)
Household head	1.490*** (0.0885)	1.418*** (0.0352)	0.838 (0.106)	0.709* (0.108)	0.656*** (0.0696)
Minority	2.585*** (0.292)	1.200*** (0.0187)	1.044 (0.0925)	2.141*** (0.366)	1.762*** (0.164)
Educ. Ref.: elementary & less ¹					
H.S. incomplete	0.553*** (0.0342)	0.620*** (0.0149)	0.749*** (0.0566)	0.319*** (0.0301)	0.732** (0.0866)
H.S. complete	0.176*** (0.0110)	0.256*** (0.00547)	0.242*** (0.0195)	0.151*** (0.0162)	0.329*** (0.0318)
Some tertiary and more	0.0742*** (0.00555)	0.107*** (0.00403)	0.0905*** (0.0124)	0.0617*** (0.00873)	0.0522*** (0.00553)
Year	0.937*** (0.00493)	0.958*** (0.00213)	0.965*** (0.00484)	1.051*** (0.00883)	0.948*** (0.00887)
Inverse Mills	1.051 (0.116)	1.481*** (0.0860)	0.529** (0.106)	0.470** (0.109)	0.379*** (0.0632)
N	91,458	154,564	34,562	14,326	13,173

Notes: (1) See [Supplement 2](#) for Complete List of Sources. (2) Exponentiated coefficients; Standard errors in parentheses. ⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. For Argentina, we clustered the standard errors at the individual level for all analyses.

motherhood decreases after accounting for informal work across all countries, and the attenuation is statistically significant for Argentina, Brazil Mexico, and Peru. Therefore, we find support for Hypothesis 3, with results suggesting that pay gaps affecting mothers are in part explained by mothers' greater likelihood to be informal workers. At the same time, this reduction is fairly modest for Brazil, so that its statistical significance could be at least partially due to the larger sample size available for this country. Furthermore, the wage gap remains virtually unaltered for Chile, a result consistent with motherhood not being a pushing factor toward informality in our previous analysis for this country. Yet,

Table 5. Effect of Motherhood on Hourly Wages (ln), from Heckman Selection Models

	Argentina, 2004–2014		Brazil, 2005–2015		Chile, 2000–2015		Mexico, 2008–2016		Peru, 2008–2016	
	Heckman	Heckman	Heckman	Heckman	Heckman	Heckman	Heckman	Heckman	Heckman	Heckman
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Mother	–0.147*** (0.0249)	–0.0815*** (0.0243)	–0.118*** (0.00398)	–0.112*** (0.00393)	–0.210*** (0.00922)	–0.206*** (0.00917)	–0.143*** (0.0202)	–0.112*** (0.0197)	–0.203*** (0.0176)	–0.175*** (0.0169)
Informal worker	–0.851*** (0.0276)			–0.249*** (0.00381)		–0.175*** (0.00889)		–0.417*** (0.0147)		–0.424*** (0.0125)
Change in coefficient from model 1 (%)	–44.56		–5.08		–1.90		–22.14			–13.64
Prob. > Chi–2	0.0000		0.0000		0.0561		0.0001		0.0001	
N	91,458		154,564		34,562		14,326			13,173

Notes: See Supplement 2 for complete list of sources. * $p > 0.05$, ** $p > 0.01$, *** $p < 0.001$ (two-tailed tests). For Argentina, we clustered the standard errors at the individual level for all analyses.

informal work explains a not-trivial part of the wage gap affecting mothers in Argentina, Mexico, and Peru.

Moreover, the reduction in our coefficient for motherhood is not uniform across the countries included. We find that informal work explains a larger share of the motherhood pay gap in Argentina versus other countries in our study. In alternative specifications of labor earnings¹⁵, the noted trend remains, although attenuated, with informal labor in Argentina showing a higher explanatory power for mothers' disadvantage. Plausibly, the strength of union organization in this country, which exclusively represents the formal workforce, may have the unintended consequence of amplifying the relative disadvantage of mothers. While working with a different strategy and period of analysis, this result further points to the trend identified by [Casal and Barham \(2013\)](#), who found a penalty in Argentina concentrated among mothers in the informal sector.

Additionally, informal work explains around a fifth of the wage gaps affecting mothers in Mexico, and around a 14 percent of that for Peru. That is, for Peru, the motherhood pay gap went down from 20 percent in our original model to 17.5 percent in our model accounting for informal work. For Mexico, the penalty was reduced from about 14 percent in Model 1 to 11 percent in Model 2. While adding a control for informal work reduces the overall motherhood wage gap, the main effect for parenthood persists statistically significant for all five countries, ranging from 8 percent in Argentina to 21 percent in Chile.

Ancillary Analyses

Accounting for occupation, part-time employment and tenure

We implemented models accounting for this set of factors, commonly thought in the causal pathway explaining mothers' disadvantage, and with tenure used as a proxy for work experience. A wage impact due to motherhood persists in these models ([Supplement 10](#)).¹⁶

Matching mothers and childless women

Results from our analyses after matching mothers and childless women are available in [Supplement 11](#), where we also present additional information on the matching procedure implemented. While slightly reduced, penalties remain significant in all countries in our study.

Taken together, our results show a significant pay gap affecting mothers, remaining even after accounting for some factors endogenous to motherhood and earnings as we did in ancillary analysis. Future research should explore explanations for this residual gap. Plausibly, discrimination may be one of them. For instance, in a study for Lima (Peru), [Galarza and Yamada \(2012\)](#) employ a laboratory experiment in order to test the prevalence of discrimination in hiring and found evidence of gender discrimination affecting women. Gender discrimination may be indicative of similar processes affecting mothers vis-à-vis childless women. Moreover, differences in productivity could also play a role at explaining the pay gap affecting mothers. The existence of such differences

is plausible, given the drastic gender division of family labor prevalent in Latin American countries (Valladolid and Lopez 2011; Pinheiro et al. 2016).

Discussion

This article provides an assessment of the wage impact of motherhood in five developing Latin American countries, paying attention to the interplay between motherhood and informal labor. We set out with two overall goals: (a) to assess the presence and magnitude of motherhood pay gaps in Argentina, Brazil, Chile, Mexico, and Peru and (b) to examine the impact of labor informality on the motherhood pay gap. Capitalizing on the richest data available for each country, our study makes the following contributions.

First, we demonstrate the presence of an important motherhood pay gap across all five countries in our study. The wage difference affecting mothers range from 12 percent in Brazil to 21 percent in Chile in our first set of models. While this range offers some variation, trends are not markedly divergent. Across a set of five Latin American countries that differ in key aspects such levels of poverty and informality, we find a similar pattern of pay disadvantage affecting mothers.

Second, our findings suggest that mothers' greater likelihood to work in the informal sector explains a portion of the motherhood pay gap in Latin America. In support of our second hypothesis, we find that mothers are more likely than childless women to join the informal sector in all countries analyzed but Chile. Moreover, in support of Hypothesis 3, controlling for informal work reduces the estimated motherhood penalty for all five countries in our study, and the difference is statistically significant for Argentina, Brazil, Mexico, and Peru. Taken together, these findings lend empirical evidence for the significance of labor informality for women in Latin America. In particular, our findings suggest that mothers are pushed toward precarious forms of employment and therefore receive lower compensations.

While our results do not present the marked contrast found for other regions (i.e., Europe), we find some variation in the wage disadvantage affecting mothers, and in the impact of labor informality at explaining pay gaps. Thus, the informal sector plays a more salient role at explaining labor outcomes of Argentinean women. We speculate that workers' collective organization may play a role in explaining this result. Unions could contribute to the protection of working mothers in the Latin American region (for Uruguay, Batthyany 2007); yet in developing countries, unions protect formal workers. Therefore, unions may contribute to the gap between mothers and childless women by channeling mothers into the informal sector in Argentina, a country with a strong tradition of collective wage bargaining (Arias et al. 2015).

On the other hand, we do not find evidence for a link between informal work and mothers' disadvantage in Chile. Plausibly, the limited participation of Chilean women in the workforce, combined with the small size of the informal sector in this country contributes to this result. That is, while in other countries, mothers may need to evaluate the opportunities and costs of formal versus

informal work, a substantial share of Chilean women may opt out from the labor market.

In this study, we highlight labor informality as a critical factor for investigating working women in developing countries. Where a rich body of Latin American research has investigated informal work, empirical contributions have largely focused on men. Yet, the processes behind women's overrepresentation among informal workers are highly gendered. Parenting is a pushing factor toward precarious work for women, but not for men. In fact, fathers are less likely to hold an informal job compared to men without children in four out of the five countries in our study, while for all countries, the coefficient of interest has the expected, negative sign.¹⁷

Once in the informal sector, mothers are likely to accumulate a low-quality work trajectory. Furthermore, the prevalence of a path connecting motherhood with job precarity is likely to reify traditional care regimes that put women as main responsible of family obligations, with their paid work taking a secondary role. It is plausible then that mothers' work in the informal sector has implications for critical issues such as their bargaining power within families and their overall chances of gaining economic independence from men.

Globally, researchers agree in that mothers' opportunities in the labor market are constrained by a number of interconnected factors; therefore, solutions are hardly straightforward. Yet, policies targeted at providing flexibility to working mothers in formal firms (e.g., in terms of schedules) may prevent them from missing opportunities in the formal sector (Heymann 2006).¹⁸ Public-funded childcare has been found positively associated with women's labor outcomes in several developed countries (Pettit and Hook 2005, 2009; Budig et al. 2016), and we would also expect that it would benefit mothers' chances of either keeping or finding a formal job (Ruel et al. 2006). Perhaps more importantly, we expect that initiatives aimed at fostering a more-egalitarian division of unpaid family labor within families could ease mothers' time constraints in Latin American countries.

Our study has limitations and presents several avenues for future research. Due to data constraints and our desire to obtain comparable results, we use cross-sectional data. Most large surveys in Latin America are not designed to follow respondents over time, and the available panel data are restricted to short periods. In this vein, our data sources do not include measures for total work experience. Future work, especially if using long-term longitudinal data, will provide valuable insights into mothers' compensation.

Moreover, informal work is indeed correlated with work-related characteristics that are negatively associated with wages (ILO 2013). We have posited that mothers' decision to look for a job in the informal sector often precedes the specific job they will find. This notion is aligned with research that acknowledges women's need for flexibility as an explanation for their overrepresentation among informal workers. It is also consistent with our analysis including work-related characteristics, where the coefficient for informality remains significant. Yet, our study does not test the trajectory from motherhood to the informal sector and then to specific jobs. Future studies, particularly using long-term

longitudinal data, will be able to document the work trajectories of women after motherhood.

Additionally, the Heckman correction we implemented is not free of valid criticism. One could argue that the variables used for identification are still connected to women's earnings. Research that proposes new strategies for addressing differential selection into employment will be critical for future work on motherhood penalties. Similarly, given we focus on wages from respondents' main job, our estimations do not consider secondary jobs. Additional studies are needed to explore the role of secondary jobs on mothers' compensation.

Another important matter refers to women self-employed and in rural areas, who are excluded from our analyses and constitute a substantial group across the region, particularly in Peru. Where ancillary analyses including these two segments do not depart drastically from our results, future work is needed to develop frameworks capable of incorporating them.

Despite these limitations, our study advances the literature on gender and labor markets. We expand upon previous work by examining the presence of motherhood wage gaps in a region for which less is known on this matter, while addressing the role of informal labor, the most-salient feature of labor markets in developing countries. In doing so, we argue that mechanisms explaining women's labor outcomes are unlikely identical across societies.

Yet, we suspect that a parallel relationship exists between precarious work and motherhood penalties in developed societies. Nonstandard work arrangements, which tend to produce precarious jobs, have risen significantly over the past 10 years in developed nations (Katz and Krueger 2016). Mothers in developed countries are also pushed toward precarious forms of employment, such as contingent work, and these movements are consequential to their work trajectories. Does precarious work exacerbate motherhood penalty? Is the penalty larger for mothers with nonstandard jobs? What is the role of self-selection? These questions warrant continued investigation.

Notes

1. Results shown in [Supplement 10](#).
2. [Supplement 1](#) provides a summary of regulations targeted to working mothers in the countries examined.
3. Sources for Argentina and Mexico (up to 2008) do not allow for identifying mothers if they are not household heads or partners. For Brazil, Chile, and Peru, we conducted analysis including three-generation households and results remain substantially stable ([Supplement 12](#)).
4. Nevertheless, our analyses include a sizeable proportion of working mothers with younger children. See [Supplement 4](#).
5. After other sample restrictions are applied, zero-income workers constitute around 2 percent or less in Argentina, Brazil and Chile, around 4 percent in Mexico, and about 7 percent in Peru. For all countries, our measure of motherhood is not associated to zero-income work (See [Supplement 13](#)).

6. Proportions of excluded observations vary according to the prevalence of self-employed and rural workers in each country. See [Supplement 3](#).
7. In most of our data sources, a more-detailed battery of questions is available for the main job. Proportions of women in our analytical samples holding more than one job range from 5 percent in Brazil to 17 percent in Peru. Only in Mexico and Peru, mothers in our analytical samples are more likely to have more than one job compared to childless women.
8. We use the official exchange rate for each year and country proportioned by the World Bank (local currency unit per US \$).
9. We evaluated each source of income corresponding the main job available in the data sources we used. Practices on this regard vary substantially by country. Each benefit, such as bonuses or payments in kind, was added to our calculation while considering its periodicity (annual, monthly, etc.)
10. Labor migrants are defined as those born in Bolivia, Colombia, Paraguay, or Peru.
11. In particular, tenure is likely to underestimate the work experience of women from low socioeconomic status, since precarious jobs are characterized by short tenures. This measure is not available for Mexico.
12. Results from Heckman selection models are shown in [Supplement 8](#).
13. We first simultaneously estimated the two models at once, using the seemingly unrelated estimation command in Stata (`suest`). This combines estimates from the two models and produces a simultaneous covariance matrix that is appropriate even when working with the same data ([Stata Corp. 2015](#)).
14. Budig and colleagues work with cross-sectional data and control for a similar set of covariates as our study, while also applying a Heckman adjustment. Their model after controlling for selection includes controls for marital status, age, education, and part-time work. While we do not control for part-time work in our main models, adding a control for part-time status slightly increases the magnitude of the penalties shown in [Table 4](#). Results are substantially similar, though.
15. We predicted monthly wages controlling for hours of work ([Supplement 18](#)). The share of the motherhood penalty explained by informal labor remains substantially larger for Argentina versus other countries, although the difference is less pronounced. The change in the coefficient for motherhood from Model 1 is of 25 percent for Argentina, 15 percent for Mexico, 14 percent for Peru, 3 percent for Brazil, and 2 percent for Chile.
16. Part-time work is associated with higher hourly wages in our analyses, a result that is commonly found in research for Latin America (e.g., [Casal and Barham \(2013\)](#); [Nopo 2012](#); [Hoyos and Nopo 2010](#)).
17. Results available upon request.
18. Indeed, this is a complex matter. In some developed countries, the use of accommodations targeted at providing flexibility to mothers has been found detrimental to their wage growth ([Glass 2004](#)).

About the Authors

Aida Villanueva is a Ph.D. candidate in sociology at the University of Texas at Austin. Her main areas of research of interest are gender and work, family labor, and the transmission of gender inequality.

Ken-Hou Lin is an associate professor of sociology department at the University of Texas at Austin. His primary research examines how the economic and demographic changes in past four decades shape the distribution of resources in the United States.

Supplementary Material

Supplementary material is available at *Social Forces* online, <http://sf.oxfordjournals.org/>.

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